

GANGNEUNG AB, Korea, Republic of

WMO: 471070

Lat: 37.754N

Lon: 128.944E

Elev: 11

StdP: 101.20

Time Zone: 9.00 (E09)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.0	-7.5	-18.9	0.7	-4.0	-16.1	0.9	-3.9	13.0	1.2	11.9	-0.2	4.5	260	0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.8	32.5	25.0	31.0	24.6	29.2	23.4	26.2	30.5	25.6	29.5	24.8	28.4	4.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.2	28.5	24.2	19.1	27.4	23.8	18.7	27.0	81.7	30.3	78.5	29.8	75.3	28.7	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
10.2	8.5	7.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			-13.3	28.6	1.5	1.2	-14.4	29.5	-15.3	30.2	-16.2	30.9	-17.3	31.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	-0.2	1.2	5.5	11.7	16.5	19.8	23.9	24.4	19.6	14.1	8.6	3.6
	DBStd	8.95	3.41	3.36	3.35	3.99	3.50	3.11	3.39	2.69	2.73	3.13	3.53	3.63
	HDD10.0	1009	318	246	145	26	1	0	0	0	0	6	66	201
	HDD18.3	2652	576	479	398	203	79	17	2	0	16	133	292	457
	CDD10.0	1906	0	0	6	76	201	295	431	448	288	134	25	3
	CDD18.3	508	0	0	0	3	22	62	175	189	54	3	0	0
	CDH23.3	3764	0	0	0	28	173	407	1425	1424	283	22	1	0
	CDH26.7	1176	0	0	0	4	31	108	511	468	52	2	0	0
Wind	WSAvg	2.8	4.1	3.2	3.3	3.3	2.8	2.2	2.3	1.9	2.0	2.3	2.9	3.7
Precipitation	PrecAvg	1403	52	54	70	79	79	111	231	277	220	107	80	39
	PrecMax	2096	248	196	174	248	277	343	758	1137	559	348	348	234
	PrecMin	924	0	0	8	1	3	14	58	55	0	2	0	0
	PrecStd	279	50	48	43	56	55	70	134	178	149	90	75	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.9	13.3	19.2	27.0	29.8	32.7	34.2	35.0	30.8	25.9	21.9	16.0
		MCWB	6.1	8.1	10.1	17.2	18.4	22.2	25.9	27.1	24.1	17.2	13.7	10.1
	2%	DB	9.0	10.2	15.9	23.2	27.2	29.8	32.2	32.9	28.1	23.1	18.6	14.0
		MCWB	4.7	5.5	8.7	14.0	17.4	20.9	25.2	25.8	22.7	16.8	12.6	8.3
	5%	DB	7.2	8.2	13.5	21.1	25.4	27.9	31.0	31.0	26.1	21.9	16.9	11.9
		MCWB	3.3	4.2	7.8	12.9	16.8	20.5	25.0	25.3	21.3	16.6	11.8	6.7
	10%	DB	5.8	6.9	11.7	19.0	23.4	25.9	29.7	29.2	24.8	20.2	15.1	10.0
		MCWB	2.3	3.6	6.8	12.0	15.9	19.8	24.2	24.5	20.7	15.5	10.4	5.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	8.9	11.2	18.4	21.0	24.0	27.0	27.7	25.2	19.8	15.1	10.8
		MCDB	10.2	11.2	16.7	23.7	26.1	29.0	31.5	33.1	29.4	22.9	18.6	14.7
	2%	WB	5.1	6.5	9.8	15.9	19.0	22.5	26.2	26.5	23.7	18.6	13.7	8.9
		MCDB	8.1	8.9	14.5	21.0	25.4	27.0	30.7	30.9	26.7	21.5	17.7	13.5
	5%	WB	3.8	4.9	8.4	14.1	17.7	21.4	25.5	25.9	22.5	17.5	12.5	7.2
		MCDB	7.0	7.4	12.2	19.2	23.5	25.6	29.8	29.5	25.3	20.5	15.9	10.6
	10%	WB	2.4	3.8	7.2	12.7	16.6	20.6	24.7	25.2	21.2	16.2	11.2	5.9
		MCDB	5.3	6.2	10.8	17.7	21.7	24.4	28.6	28.6	23.8	19.5	14.4	9.6
Mean Daily Temperature Range	5% DB	MDBR	9.3	8.3	8.9	9.9	9.4	7.2	6.5	6.8	8.8	10.7	10.3	9.2
		MCDBR	10.4	10.3	12.0	12.8	13.2	11.0	9.2	9.9	10.3	12.4	12.5	11.3
		MCWBR	7.9	7.3	7.8	7.9	7.5	5.7	4.7	4.8	6.3	8.1	8.5	7.9
	5% WB	MCDBR	10.2	9.1	11.0	11.4	11.5	9.2	8.2	8.4	8.9	10.8	11.7	10.3
		MCWBR	8.3	6.9	7.9	7.9	7.3	5.6	4.8	4.7	6.0	7.9	8.6	7.9
Clear-Sky Solar Irradiance	taub	0.422	0.487	0.584	0.625	0.630	0.680	0.641	0.571	0.498	0.490	0.454	0.413	
	taud	1.966	1.842	1.681	1.651	1.684	1.647	1.788	1.962	2.088	1.978	1.997	2.019	
	Ebn at Noon	709	711	681	686	694	659	682	722	753	702	672	687	
	Edh at Noon	135	175	226	246	242	251	216	178	148	150	131	121	
All-Sky Solar Radiation	RadAvg	2.39	3.17	4.11	4.91	5.43	5.16	4.31	4.28	3.90	3.40	2.41	2.14	
	RadStd	0.20	0.30	0.40	0.38	0.53	0.44	0.61	0.53	0.47	0.30	0.31	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (34 neighbors)	N/A	N/A	-0.94	N/A	N/A	N/A	N/A	N/A	+97	+52

Nomenclature: See separate page